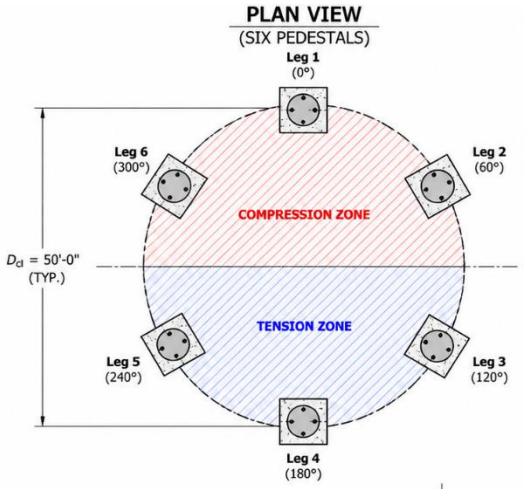
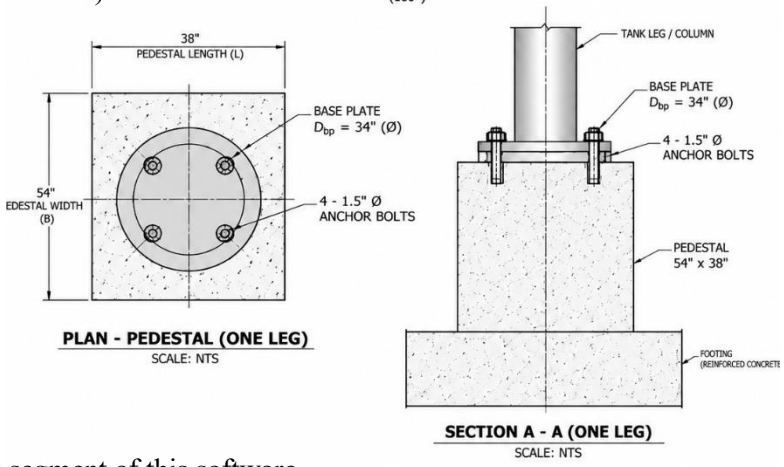
3: RINGWALL Foundation Design:

Design Parameter	Symbol	Value	Source
Total factored axial load	P_u	**4623.2018 kips**	Main Tank Design (Predefined)
Governing factored overturning moment	M_u	**3823.8 kip-ft**	Main Tank Design (Predefined)
Total factored horizontal shear	V_u	**57.218 kips**	Main Tank Design (Predefined)
Number of legs	N_{leg}	**6**	User Input
Diameter circle representing axisymmetric spacing between legs/columns	D_{cl}	**50'-00"**	Main Tank Geometry (Predefined)
Pedestal size	$B \times L$	**54 in × 38 in**	User Input
Base plate size	D_{bp}	**34 in**	User Input
Concrete compressive strength	f_c'	**4.0 ksi**	User Input / ACI 318-19
Reinforcing steel yield strength	f_y	**36 ksi**	User Input / ACI 318-19
Anchor bolts	$N_b \times d_b$	**4 × 1.5 in**	Designed in Anchor Bolts segment of this software
Anchor embedment depth	h_{ef}	**44 in**	Designed in Anchor Bolts segment of this software
Pedestal clear height used	H_p	**≈ 8'-6" = 102 in**	User Input
Concrete unit weight	γ_c	**150 pcf**	User Input





Step 1-2: ACI 318-19 slenderness check

ACI 318-19 6.2.5.1 permits slenderness effects to be neglected for a column not braced against sidesway when:

For a square section:

$$r = \sqrt{\frac{I}{A}}$$

I = moment of inertia of the section

r = Radius of gyration

A = Sectional Area

$$I_x = \frac{bh^3}{12} = \frac{54(38)^3}{12} = 246924 \text{ in}^4$$
$$I_y = \frac{hb^3}{12} = \frac{38(54)^3}{12} = 498636 \text{ in}^4$$

And:

$$A_g = 54(38)$$
$$A_g = 2052 \text{ in}^2$$

$$r_{min} = \sqrt{\frac{246924}{2052}}$$

r = 10.970 inch

Check A — Pedestal classification

$$\frac{\ell_u}{b} \leq 3$$
$$1.648 < 3$$

Therefore, it qualifies geometrically as a pedestal.

Check B — Structural Slenderness

$$\frac{k\ell_u}{r} \leq 22$$

Using the conservative $k = 2.0$:

$$16.227 < 22$$

Therefore, second-order slenderness effects can be neglected.

Step 3: Gravity axial load per leg

For four equally spaced legs:

$$P_{gravity} = \frac{P_u}{N_{leg}}$$
$$P_{gravity} = \frac{4623.2}{4}$$
$$\boldsymbol{P_{gravity} = 770.534 \text{ kip/leg}}$$

The self-weight pedestal can also be included.

γ_c=150 pcf

$$W_p = (54/12)(38/12)(102/12)(0.150)/1000 \text{ kips}$$
$$W_p \approx 18.169 \text{ kips}$$

Factored:

$$1.2W_p = 21.803 \text{ kips}$$

Therefore, conservatively:

$$P_{comp,total} = 770.534 + 21.803$$
$$\boldsymbol{P_{comp,total} = 792.336 \text{ kip}}$$

Step 4: Pedestal Concrete Bearing Check

Base plate:

$$A_1 = \frac{\pi(34)^2}{4} = 907.920 \text{ in}^2$$

Pedestal:

$$A_2 = 54(38) = 2052 \text{ in}^2$$

Bearing enhancement factor:

$$\sqrt{\frac{A_2}{A_1}} = \sqrt{\frac{2052}{907.920}} = 1.503$$

Using the concrete bearing-strength relationship: $\phi P_n = \phi(0.85 f_c' A_1)\sqrt{\frac{A_2}{A_1}}$

Taking:

Transverse Pedestal Reinforcement: if

1. Spiral = 0.75
2. Ties = 0.65

$$\phi P_n = 0.65(0.85)(4)(907.920)(1.503)$$
$$\boldsymbol{\phi P_n \approx 3016.510 \text{ kips}}$$

Demand:

$$P_u = \mathbf{792.336} \text{ kip}$$
$$\frac{\boldsymbol{Pmax}}{\boldsymbol{\phi P_n}} = \frac{\mathbf{792.336}}{\mathbf{3016.510}} = 0.28$$

Therefore:

$0.28 < 1.00$……. **CONCRETE BEARING — PASS**

Demand/capacity ratio:……. $D/C \approx 0.262$

So the **39 × 39 in pedestal is more than adequate for bearing**.

Step 5: Pedestal longitudinal Reinforcement

$18 - \#6$ vertical bars

Area of one #6:

$$A_b = 0.44 \text{ in}^2$$

$$\rho_g = \frac{A_s}{A_g} = \frac{7.92}{2{,}052} = 0.00386$$
$$\rho_g = 0.386\%$$

$$A_{s,min} = 0.005 A_g$$

ACI 318-19 Section 16.3.4.1 specifies the $0.005 A_g$ minimum for connections between a cast-in-place column or pedestal and its foundation.

As,min=0.001(2052)

$$A_{s,min} = 10.26 \ in^2$$

Provided:

$$A_s = 7.92 \text{ in}^2$$

Therefore:

$7.92 > 7.605$…. FAIL - under this minimum-reinforcement requirement

Therefore, based on the longitudinal reinforcement in the drawing:

Based on the drawing 14 #8 vertical bars.
The area of one #8 bar is 0.79in^2.

$$A_s = 14(0.79) = 11.06\text{in}^2$$

Reinforcement ratio

$$\rho_g = \frac{A_s}{A_g} = \frac{11.06}{2{,}052} = 0.00539$$
$$\rho_g = 0.539\%$$

Minimum reinforcement check
Using the $0.005A_g$ minimum assumed in the earlier calculation, where that provision applies:

$$A_{s,\text{min}} = 0.005(2{,}052) = 10.26\text{in}^2$$

Compare:

$$\mathbf{11.06 > 10.26 \ldots . Pass}$$

Step 6: Axial Compression Capacity

$f_c' = 4.0\text{ksi}$

$f_y = 36\text{ksi}$.
For the illustrative concentric nominal axial strength:

$$P_o = 0.85f_c'(A_g - A_s) + f_y A_s$$
$$P_o = 0.85(4.0)(2{,}052 - 7.92) + 36(7.92)$$
$$P_o = 6{,}949.872 + 285.120$$
$$P_o = 7{,}234.992\text{kips}$$

For a tied member, using the illustrative maximum axial-strength limit of $0.80P_o$ and $\phi = 0.65$:

$$\phi P_{n,\text{max}} = 0.65(0.80)(7{,}234.992)$$
$$\phi P_{n,\text{max}} = 3{,}762.2\text{kips}$$

Using $P_{\text{max}} = 792.336$ Kips :

$$D/C = \frac{792.336}{3{,}762.2} = 0.212$$

Step 7: Pedestal Tensile Capacity Check

Using the foundation drawing's 14 #8 vertical bars:

$$A_s = 14(0.79) = 11.06\text{in}^2$$

Using your specified reinforcing steel yield strength, $f_y = 36$ksi, and assuming $\phi = 0.90$:

$$\phi T_n = \phi A_s f_y$$
$$\phi T_n = 0.90(11.06)(36)$$
$$\phi T_n = 358.344\text{kips}$$

Calculate the demand/capacity ratio:

$$D/C = \frac{T_u}{\phi T_n} = \frac{114.6925}{358.344}$$
$$D/C = 0.320$$

Since:

$114.6925 < 358.344$kips…….PASS - steel tensile capacity check.

Step 8: Pedestal *P-M* Interaction Check

Parameter	Value
Section width	54 in
Section depth	38 in
Concrete compressive strength	4.0 ksi
Reinforcing steel yield strength	36 ksi
Gross concrete area	2,052in^2
Longitudinal reinforcement	14 #8 bars
Total longitudinal steel area	11.06in^2

Design section properties

Steel yield strain

$E_s = 29{,}000$ksi

$$\epsilon_y = \frac{f_y}{E_s} = \frac{36}{29{,}000}$$
$$\epsilon_y = 0.001241$$

Balanced neutral-axis depth

The balanced neutral-axis depth

$$c_b = d\frac{0.003}{0.003 + \epsilon_y}$$

Substituting the steel yield strain:

$$c_b = d\frac{0.003}{0.003 + 0.001241}$$
$$c_b = 0.7073d$$

Equivalent compression-block depth

For $f_c' = 4.0\text{ksi}$, the applicable ACl stress-block factor must be confirmed for the governing code provisions. If $\beta_1 = 0.85$ applies:

$$a_b = \beta_1 c_b$$
$$a_b = 0.85(0.7073d)$$
$$a_b = 0.6012d$$

Step 9: Pedestal Tie Design

9.1 Tie diameter

ACI 318-19 Section 25.7.2.2 specifies the minimum tie-bar diameter based on longitudinal bar size:

- No. 3 ties for No. 10 or smaller longitudinal bars.
- No. 4 ties for No. 11 or larger longitudinal bars, or bundled longitudinal bars.

The foundation drawing specifies 14 No. 8 vertical bars. Therefore:

$$d_{bt} \geq \text{ No. 3}$$

For No. 3 ties:

$$d_{bt} = 0.375\text{in}$$

However, because the drawing specifies No. 4 hoops, use No. 4 hoops in the final design unless an approved revision states otherwise.

9.2 Maximum tie spacing

ACI 318-19 Section 25.7.2.1 gives the following spacing limits for the applicable tied-member configuration:

$$s_{\max} = \min(16d_b, 48d_{bt}, b_{\min})$$

For the specified No. 8 longitudinal bars:

$$d_b = 1.00\text{in}$$

For the minimum No. 3 tie:

$$d_{bt} = 0.375 \text{ in}$$

The minimum pedestal dimension is:

$$b_{\min} = 38 \text{ in}$$

Calculate each limit:

$$16d_b = 16(1.00) = 16\text{inch}$$

$$48d_b = 48(0.375) = 18\text{inch}$$

$$b_{\min} = 38 \text{ in}$$

$$s_{\max} = \min(16,\ 18,\ 38)$$

$$s_{\max} = 16 \text{ in}$$

Adopt tie spacing

A preliminary spacing selection is:

$$s = 12 \text{ in}$$

$$12 < 16 \text{ in}$$

Thus, the selected spacing satisfies the calculated maximum-spacing limit.

Step 10: Pedestal-to-Footing Development

ACI 318-19 Section 25.4 provides the development and anchorage requirements for reinforcing bars. Because the foundation drawing specifies hooked vertical bars, the final calculation must use the applicable hooked-bar provisions if standard hooks are actually provided.

The following calculation documents the straight-bar development-length expression from your supplied calculation. It is not a substitute for the required hooked-bar check.

parameters

Parameter	Symbol	Value
Concrete compressive strength	f'c	4.0 ksi
Reinforcing steel yield strength	fy	60 ksi, as assumed in the supplied calculation
Lightweight concrete factor	λ	1.0
Top-bar factor	ψ_t	1.0, if applicable
Coating factor	ψ_t	1.0, if applicable
Bar-size factor	ψ_t	1.0 for No. 8 bars
Steel-grade factor	ψ_t	1.0 for Grade 60
Assumed confinement ratio	cb+K_{tr}/db	2.5
Bar diameter	d_b	1.00 in for a No. 8 bar

Calculate straight-bar development length

$$l_d = \frac{3}{40}\frac{f_y}{\lambda\sqrt{f'_c}}\frac{\psi_t\psi_e\psi_s\psi_g}{(c_b + K_{tr})/d_b}d_b$$

$$l_d = \frac{3}{40}\frac{60{,}000}{1.0\sqrt{4{,}000}}\frac{1.0(1.0)(1.0)(1.0)}{2.5}(1.00)$$

$$l_d = 28.46 \text{ in}$$

$$l_{\text{adopted}} = 30 \text{ in}$$

$30 > 28.46$ in

Preliminary straight-bar length check: PASS